

Soham Kulkarni

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Education

University of California, Los Angeles (UCLA)

Sep 2023 – June 2025

Master of Science (MS) in Computer Science

Indian Institute of Technology (IIT) Hyderabad

Aug 2019 – May 2023

Bachelor of Technology in Electrical Engineering (Minor in Entrepreneurship)

Publications & Preprints

- **S. Kulkarni**, R. Dhar, Y. Cui. *Learning from the Best: Smoothness-Driven Metrics for Data Quality in Imitation Learning*. ICML 2025 DataWorld Workshop, Vancouver, Canada, July 2025. [\[paper\]](#) (Also submitted to IROS '26)

Technical Skills

Programming: Python, C/++, MATLAB · **Software/Tools:** Git, Docker, ROS/ROS2, CUDA, wandb, roboflow, SolidEdge, Autodesk Fusion · **Libraries:** PyTorch, TensorFlow, OpenCV, scikit-learn, pandas, TensorRT, robosuite, HuggingFace, ZED/Realsense SDK · **Simulators:** MuJoCo, IssacSim, Unity, Gazebo, rViz, PyBullet, AirSim

Research Experience

Benchmarking Data Quality in Imitation Learning [M.S. Thesis]

Los Angeles, CA

Under Dr. Yuchen Cui, UCLA Robot Intelligence Lab (URIL)

Sep 2024 – Ongoing

- Developed RINSE, an offline trajectory-quality framework for robot imitation learning using two smoothness metrics: **TED** (contact-aware Cartesian trajectory deviation) and **SAL** (frequency-domain smoothness).
- Improved data efficiency: **+16%** RoboMimic Transport success using only **50/300** demonstrations (**6×** less data), and **+20%** success on a real-world Push-Block task using half the data (**+16%** with one-fourth data); filtering outperformed random subsets by **8–40 pp**.
- Filtering + Retrieval-Augmented Learning: **+5.6%** mean success as a retrieval signal in STRAP on LIBERO-10; metrics show task-dependent complementarity (TED on contact tasks: Cream-Cheese-Butter **+32%**, SAL on free-space motion: Book-Caddy **+20%**) and align with learned Re-Mix domain weights (Spearman $\rho \geq 0.89$).
- Validated across data-collection modalities (kinesthetic > VR > SpaceMouse); trajectory scores correlate with downstream policy success and operator-reported task difficulty.
- Provided theoretical analysis showing smoothness filtering reduces conditional action variance in behavioral cloning, explaining improvements under compounding error and action chunking.

Stereo Vision and Planning for the ARTEMIS Humanoid [RoboCup 2024 Winner]

Eindhoven, Netherlands

Under Dr. Dennis Hong, RoMeLa, UCLA

Jan 2024 – Jul 2024

- Created a “proximity” package for near-field planning at 60 Hz using connected-components analysis + union-find on stereo depth maps to detect the closest obstacles.
- Extracted obstacle features (width, height, distance, angle) and transformed them into robot and field reference frames for use by localization and mid-level planning modules.
- Deployed adaptive depth thresholding to suppress ground/self detections and designed a HALO fallback mode for safe (and consistent) navigation under occlusions and clutter.

Bimanual Robot Manipulation on AgileX Cobot Magic

Santa Clara, CA

- Developed a ROS-based teleoperation stack (Piper SDK) with 15 Hz control, multi-camera feedback, and joint state monitoring for a bimanual setup.
- Collected teleoperated demonstrations and trained Diffusion Policy on visual and proprioceptive inputs for table-top manipulation tasks.
- Deployed 16-step vision-based closed-loop inference with ~ 2.1 s real-time inference, debugging stability issues on real hardware.

Humanoid Fall Recovery Controller (in MuJoCo)

Los Angeles, CA

- Designed hierarchical FSM architecture with capture-point based reflex cascade (ANKLE→HIP→STEP) achieving 62% standing/38% balancing under 500N perturbations with <200 Nm torque peaks.

- Implemented full-body dynamic balance using five coordinated strategies (ankle, hip, arms, torso, stepping) with CP-error gating across standing/balancing/stepping/emergency states.
- Developed trajectory-free fall mitigation pipeline with impact absorption, tripod recovery, and scripted stand-up sequences; validated against encoder dropout, terrain compliance, and random-walk disturbances up to 0.35m CP deviation.

Reinforcement Learning for Autonomous Navigation on Uncertain Terrain

Under Dr. M. Vidyasagar, FRS (IIT Hyderabad)

Hyderabad, India

Jun 2021 – May 2022

- Modeled UGV navigation on unstructured, uneven terrain as a DeepRL task and implemented an end-to-end A3C pipeline in simulation (Gazebo).
- Integrated user-defined obstacle placement and 3D occupancy mapping; implemented a dynamic obstacle scheduler during training to encourage early exploration.
- This curriculum design led to a **14%** increase in navigation success rate for a Pioneer mobile robot.

Agile Autonomous Quadrotor Flight [For IROS 2022 Safe Robot Learning Competition]

Under Dr. M. Vidyasagar (IIT Hyderabad) & Dr. Srikanth Saripalli (Texas A&M)

Hyderabad, India

Aug 2022 – May 2023

- Deployed a PPO-based controller in SE(3) for minimum-time navigation on a Crazyflie quadrotor (pycrazyswarm) under gate and obstacle constraints.
- Designed a reward structure balancing progress, lateral deviation, and collision penalties for safe and smooth flight.
- Demonstrated improved reliability under adaptive control settings with **84%** success, compared to non-adaptive baselines.

Data-Acquisition Robot for Acoustic Sensing & Localization [Robot Audition]

Under Dr. Sumohana Chhannappayya & Dr. K. Sri Rama Murty (IIT Hyderabad)

Hyderabad, India

Jul 2021 – May 2022

- Developed an all-terrain robot (height up to 1.7m) with an adjustable 8-channel microphone array for 3D sound source localization experiments.
- Implemented robust sound source localization and tracking, streaming multi-channel audio and state estimation over 4G to on-board storage for long-horizon data collection.

Select Projects

Creating Persona Chatbots (with a focus on evaluation) [For CS263 NLP @ UCLA]

Los Angeles, CA

- Used in-context and chain-of-thought prompt tuning on Mistral 7B for persona-aligned open-ended dialogues.
- Evaluated with LLMEval (fluency/coherency), MMLU (reasoning), and HDS-style metrics (toxicity/fairness) to refine persona fidelity and control harmful outputs.

CRAG Meta KDD Competition [Best Project for CS245 Big Data Analytics @ UCLA]

Los Angeles, CA

- Built an enhanced RAG system with recursive chunking, dense retrieval and reranking (BAAI/bge-reranker-v2-m3) to improve document selection and mitigate hallucinations.
- Classified queries by temporal sensitivity (static, slow-changing, fast, real-time) to adapt retrieval strategy.
- Increased open-ended query accuracy to **42.16%** using GPT-4o-mini with chain-of-thought prompting.

Effect of Neural Network Pruning on Spurious Correlations [For CS260D Large-Scale ML @ UCLA]

Los Angeles, CA

- Investigated unstructured magnitude pruning effects on neural nets with spurious correlations, comparing supervised (LeNet + JTT) and contrastive learning on SpuCo MNIST.
- Demonstrated last-layer pruning maintains worst-group accuracy up to 70% pruning ratio, outperforming early-layer pruning strategies for fairness-preserving model compression.
- Achieved 83.5% validation accuracy with 90% last-layer pruning in contrastive settings while minimizing spurious feature reliance.

Motion Planning & Teleoperation Interface [For Netflix's *The Electric State*: Cosmo Humanoid]

Los Angeles, CA

Los Angeles, CA

- Developed a 3D planning interface for safe trajectory recording, replay, and teleoperation using Unity and PyBullet.

SLAM on Victoria Park Dataset [IEEE Summer School: SLAM in Deformable Environments]

Australia

Sydney, Australia

- Implemented EKF-based SLAM and non-linear least-squares (graph-based) SLAM on the Victoria Park dataset, comparing trajectory and landmark estimation quality.

Flipkart Grid 3.0 Robotics Challenge [National Semifinalists]

Hyderabad, India

- Led a team to develop overhead vision-based swarm robots for pick-and-place and warehouse automation tasks, including multi-robot coordination and task assignment. We were national semifinalists among 9000+ participants.

DDP-based Solver for Optimal Control for the TALOS Humanoid

Los Angeles, CA

- Implemented Differential Dynamic Programming (DDP) solvers (in simulation) for whole-body manipulation, gait planning, and bipedal walking under contact constraints.

Teaching & Mentoring

- *UCLA*: Reader for CS 217A (Internet Architecture and Protocols) and CS 162 (Natural Language Processing); conducted office hours and created assignments/quizzes for classes of ~150 students.
- *IIT Hyderabad*: Coordinator and mentor of the Robotics Club; participated in Inter-IIT and other national level competitions, mentored students on robotics projects, and conducted hands-on hardware and software sessions.

Honors & Awards

- IIT-JEE Advanced All India Rank **2362** among **190,000** shortlisted students from IIT-JEE Mains
- IIT-JEE Mains: **99.5** percentile among **1.1 million** students
- **SA Gold Medal for Innovation** at IIT Hyderabad (Institute-wide, 1 recipient among all UG students) (Apr 2023)
- Robocup Humanoids league **winner** for the Robosoccer challenge (2024)
- Flipkart Grid 3.0 Robotics Challenge: **National Semifinalist** (Team leader for IIT Hyderabad team)
- Awarded the BUILD funding grant for student-led project development (2022).
- SPIRSE Award by IEEE RAS (invited to ICRA 2022)